D44 | Build a fund, not a story

Ten-year fictional fund model | All money in INR crore | 13 November 2026 | Editable inputs are yellow; formulas are purple.

Rebuild first	Start from the D44 exercise on a blank sheet before consulting these worked formulas. This workbook is an authored check, not evidence that an independent learner has built the model.
What is modeled	Time-0 LP call of 159 funds investment cost 120 plus fee reserve 39. Commitment 300 is not paid-in capital. All exits distribute immediately; fees come from the reserve, not new calls.
Two carry calculations	Paid carry uses cumulative distributions above paid-in capital. Interim accrued carry uses cumulative distributions plus gross NAV; deduct only its unpaid portion from NAV. Do not subtract paid carry again.
Holdings and timing	Thirty investments cost 4 each. Six zeros are written off in year 6; eighteen 2x, four 7x and two 25x holdings exit on the stated schedule. Before exit/write-off, marked value equals cost by stipulation.
IRR snapshots	Each year IRR uses -159 at time 0, actual LP flows through that year and terminal net NAV added once in the measurement year. It is an interim valuation estimate, not a realized return or equal-spaced monthly XIRR.
Run the downside	On Inputs, change Winner 2 multiple from 25 to 0. Only one of the two winners changes, still in year 10. Explain proceeds, carry, net DPI, IRR and the TVPI trough; then restore the baseline.
Boundaries	This is a fixed fictional schedule with no tax, debt, follow-on, recycling or hurdle/catch-up. A modified fee/carry rule, payment timing, NAV convention or inadequate fee reserve needs a different model.
Native evidence	Recalculation and changed-case tests in LibreOffice establish authored behavior only. They do not establish Excel/Sheets parity, actual fund performance or learner transfer. No account, upload, submission or tracking is created.

Metrics are on the net LP basis unless labeled gross portfolio. Ratio and percentage-return charts are separate. Source boundaries and full classroom assumptions remain in D44.

Inputs | One controlled change

The baseline follows D44 exactly. Edit Winner 2 multiple only for the requested sensitivity; other assumptions are displayed so the basis is explicit.

Commitment	300.00
Time-0 LP paid-in	159.00
Portfolio investment cost	120.00
Number of investments	30.00
Years 1-5 annual fee rate	2.00%
Years 6-10 annual fee rate	1.50%
Years 6-10 fee cost base	120.00
Whole-fund carry rate	20.00%
Winner 2 multiple	25.00

Fee reserve at time 0	39.0000
Total scheduled fees	39.0000
Uncalled commitment	141.0000
Cost per investment	4.0000
Fee reserve less total fees	0.0000

Baseline fee reserve equals scheduled fees: 39. Negative reserve coverage invalidates this no-new-calls schedule; return metrics are then suppressed. Other changes may need a new timing, legal or economic model.

Holdings | Cost, exits and write-offs

One row per investment. Before the specified exit or write-off, value equals original cost by classroom convention. No real fair-value policy is implied.

Holding	Outcome group	Original cost	Gross multiple	Exit/write-off year	Gross proceeds	Timing check
1	Zero	4.0000	0	6	0.0000	1.0000
2	Zero	4.0000	0	6	0.0000	1.0000
3	Zero	4.0000	0	6	0.0000	1.0000
4	Zero	4.0000	0	6	0.0000	1.0000
5	Zero	4.0000	0	6	0.0000	1.0000
6	Zero	4.0000	0	6	0.0000	1.0000
7	2x - year 6	4.0000	2	6	8.0000	1.0000
8	2x - year 6	4.0000	2	6	8.0000	1.0000
9	2x - year 6	4.0000	2	6	8.0000	1.0000
10	2x - year 6	4.0000	2	6	8.0000	1.0000
11	2x - year 6	4.0000	2	6	8.0000	1.0000
12	2x - year 6	4.0000	2	6	8.0000	1.0000
13	2x - year 7	4.0000	2	7	8.0000	1.0000
14	2x - year 7	4.0000	2	7	8.0000	1.0000
15	2x - year 7	4.0000	2	7	8.0000	1.0000
16	2x - year 7	4.0000	2	7	8.0000	1.0000
17	2x - year 7	4.0000	2	7	8.0000	1.0000
18	2x - year 7	4.0000	2	7	8.0000	1.0000
19	2x - year 8	4.0000	2	8	8.0000	1.0000
20	2x - year 8	4.0000	2	8	8.0000	1.0000
21	2x - year 8	4.0000	2	8	8.0000	1.0000
22	2x - year 8	4.0000	2	8	8.0000	1.0000
23	2x - year 8	4.0000	2	8	8.0000	1.0000
24	2x - year 8	4.0000	2	8	8.0000	1.0000
25	7x - year 8	4.0000	7	8	28.0000	1.0000
26	7x - year 8	4.0000	7	8	28.0000	1.0000
27	7x - year 9	4.0000	7	9	28.0000	1.0000
28	7x - year 9	4.0000	7	9	28.0000	1.0000
29	Winner 1	4.0000	25	10	100.0000	1.0000
30	Winner 2	4.0000	25.0000	10	100.0000	1.0000
Totals		120.0000			456.0000	

The downside changes holding 30 only. A multiple of zero keeps its year-10 write-off timing; it does not retrospectively erase earlier cost-basis NAV.

Cash flow | From gross to LP

All money in INR crore. Cumulative carry is paid only as exits create distributions; NAV accrual is computed separately in Metrics.

Year	Annual fee	Gross exit	Fee cash left	Holding NAV	Gross NAV	Cum. gross exit	Cum. paid carry	Annual carry	LP distribution	Cum. LP dist.	Cash check
0	0.0000	0.0000	39.0000	120.0000	159.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	6.0000	0.0000	33.0000	120.0000	153.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	6.0000	0.0000	27.0000	120.0000	147.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	6.0000	0.0000	21.0000	120.0000	141.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4	6.0000	0.0000	15.0000	120.0000	135.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5	6.0000	0.0000	9.0000	120.0000	129.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
6	1.8000	48.0000	7.2000	72.0000	79.2000	48.0000	0.0000	0.0000	48.0000	48.0000	0.0000
7	1.8000	48.0000	5.4000	48.0000	53.4000	96.0000	0.0000	0.0000	48.0000	96.0000	0.0000
8	1.8000	104.0000	3.6000	16.0000	19.6000	200.0000	8.2000	8.2000	95.8000	191.8000	0.0000
9	1.8000	56.0000	1.8000	8.0000	9.8000	256.0000	19.4000	11.2000	44.8000	236.6000	0.0000
10	1.8000	200.0000	0.0000	0.0000	0.0000	456.0000	59.4000	40.0000	160.0000	396.6000	0.0000

Cash check is opening paid-in minus portfolio investment cost minus cumulative fees minus remaining fee cash. Gross exits pass straight to gross distributions; carry divides them between LP and manager, not fee-reserve cash.

Metrics | Do not subtract carry twice

Net LP ratios use paid-in 159. Interim IRR adds net NAV once to actual LP cash flows through the measurement year. Invalid fee reserve suppresses ratios/IRR.

Year	Gross NAV	Accrued carry	Paid carry	Unpaid carry	Net NAV	DPI (net)	RVPI (net)	TVPI (net)	IRR annual	TVPI identity	Net LP value
0	159.0000	0.0000	0.0000	0.0000	159.0000	0.0000	1.0000	1.0000	#N/A	0.0000	159.0000
1	153.0000	0.0000	0.0000	0.0000	153.0000	0.0000	0.9623	0.9623	-3.77%	0.0000	153.0000
2	147.0000	0.0000	0.0000	0.0000	147.0000	0.0000	0.9245	0.9245	-3.85%	0.0000	147.0000
3	141.0000	0.0000	0.0000	0.0000	141.0000	0.0000	0.8868	0.8868	-3.93%	0.0000	141.0000
4	135.0000	0.0000	0.0000	0.0000	135.0000	0.0000	0.8491	0.8491	-4.01%	0.0000	135.0000
5	129.0000	0.0000	0.0000	0.0000	129.0000	0.0000	0.8113	0.8113	-4.10%	0.0000	129.0000
6	79.2000	0.0000	0.0000	0.0000	79.2000	0.3019	0.4981	0.8000	-3.65%	0.0000	127.2000
7	53.4000	0.0000	0.0000	0.0000	53.4000	0.6038	0.3358	0.9396	-0.93%	0.0000	149.4000
8	19.6000	12.1200	8.2000	3.9200	15.6800	1.2063	0.0986	1.3049	3.72%	0.0000	207.4800
9	9.8000	21.3600	19.4000	1.9600	7.8400	1.4881	0.0493	1.5374	5.82%	0.0000	244.4400
10	0.0000	59.4000	59.4000	0.0000	0.0000	2.4943	0.0000	2.4943	11.43%	0.0000	396.6000

Gross portfolio MOIC	3.8000
Net terminal DPI	2.4943
Lowest post-call TVPI	0.8000
First trough year	6.0000
Terminal LP distributions	396.6000
Terminal paid carry	59.4000

Trough definition: minimum end-year net TVPI over years 1-10, using first year if tied. This is not a claimed universal J-curve trough. A ratio trough and an IRR trough can differ.

IRR snapshots | Truncate the horizon

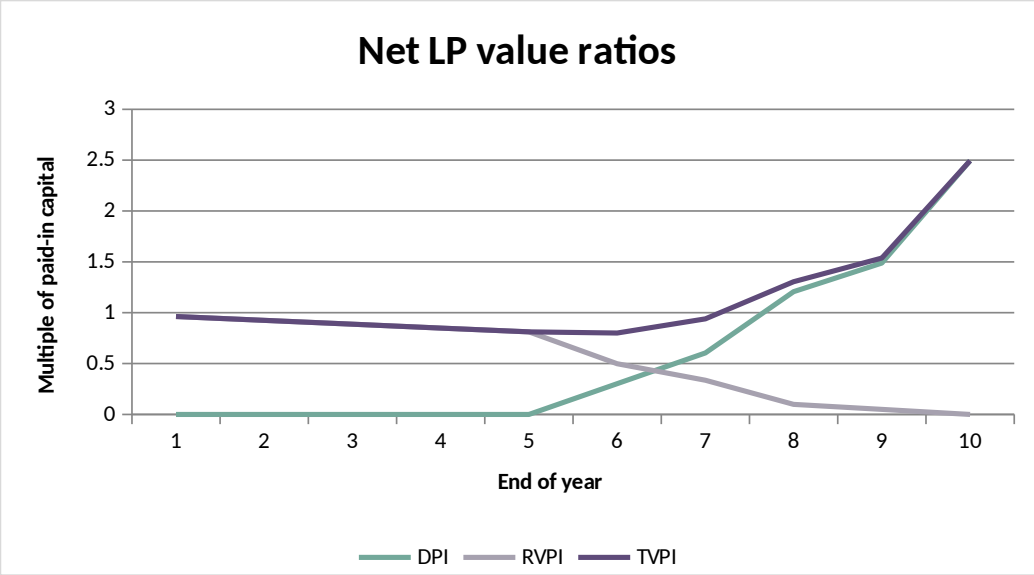
Each column is a separate valuation date. Rows after that date are excluded from IRR, not included as extra zero-flow years. Time-0 call is negative.

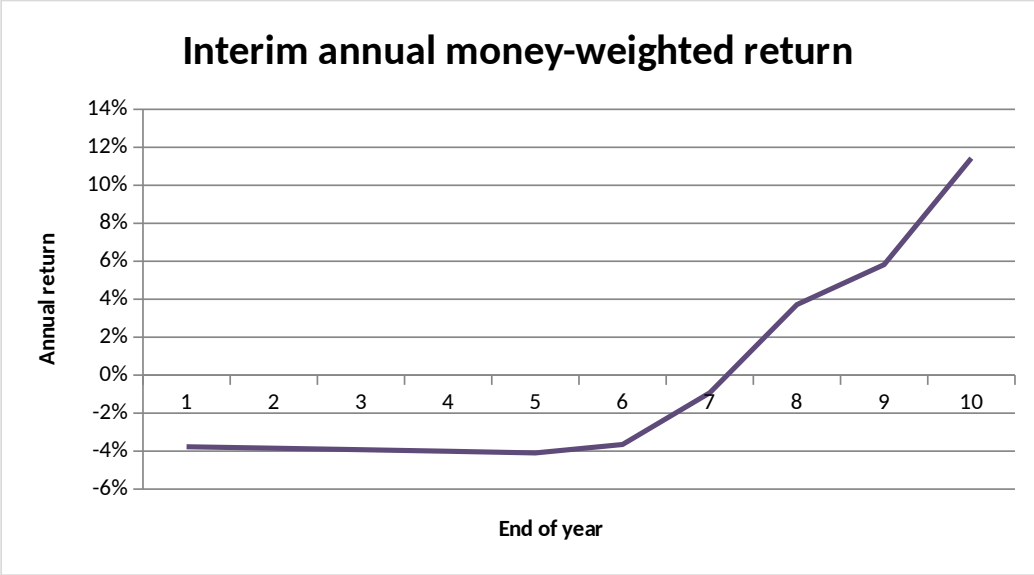
Cash-flow year	As of Y1	As of Y2	As of Y3	As of Y4	As of Y5	As of Y6	As of Y7	As of Y8	As of Y9	As of Y10
0	-159.0000	-159.0000	-159.0000	-159.0000	-159.0000	-159.0000	-159.0000	-159.0000	-159.0000	-159.0000
1	153.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0000	147.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0000	0.0000	141.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4	0.0000	0.0000	0.0000	135.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5	0.0000	0.0000	0.0000	0.0000	129.0000	0.0000	0.0000	0.0000	0.0000	0.0000
6	0.0000	0.0000	0.0000	0.0000	0.0000	127.2000	48.0000	48.0000	48.0000	48.0000
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	101.4000	48.0000	48.0000	48.0000
8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	111.4800	95.8000	95.8000
9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	52.6400	44.8000
10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	160.0000

An interim NAV estimate is not a cash distribution. It appears once only to estimate interim money-weighted return; DPI uses actual distributions only. No regular annual-rate guess substitutes for the actual sequence.

Two views | Ratios are not rates

Net LP DPI, RVPI and TVPI in multiples of paid-in; annual IRR in a separate percentage chart. Charts recalculate with inputs.





Earlier-year IRRs include terminal net NAV estimates; year 10 is fully realized in the baseline. No smooth path, fixed early trough or realized return guarantee is assumed.

Predict, rebuild, reconcile

Blank response cells are for your own work. Baseline targets below stay fixed when you change the model. Self-check only, not automatic grading.

Baseline target	Your answer	Check value	Difference
Total fees		39.0000	
Gross terminal proceeds		456.0000	
Gross portfolio MOIC		3.8000	
Terminal carry paid		59.4000	
Terminal LP distributions		396.6000	
Terminal net DPI		2.4943	
Terminal RVPI		0.0000	
Net TVPI trough		0.8000	
First trough year		6.0000	

Downside: change only Winner 2 multiple to zero. Predict and explain gross proceeds 356, carry 39.4, LP distributions 316.6 and net DPI 316.6/159. A year-10 write-off does not erase earlier cost-basis NAV.

Explain the difference between commitment, paid-in, investment cost and NAV; why unpaid rather than all accrued carry reduces NAV; why a 25x holding is not a whole-fund 25x return; and why IRR needs dates and actual flows.

Rebuild a no-carry case and a fee-reserve shortfall as diagnostic checks. A negative reserve requires funding changes; this workbook does not invent extra calls or borrowing. Do not treat suppressed returns as zeros.